

HalluScan: A Systematic Benchmark for Detecting and Mitigating Hallucinations in Instruction-Following LLMs

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Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse natural language processing tasks, yet they remain susceptible to hallucinations—generating content that is factually incorrect, unfaithful to provided context, or misaligned with user instructions. We present HALLUSCAN, a comprehensive benchmark framework that systematically evaluates hallucination detection and mitigation across 72 configurations spanning 6 detection methods, 4 open-weight model families, and 3 diverse domains. We introduce three key contributions: (1) HALLUSCORE, a novel composite metric that achieves a Pearson correlation of $r = 0.41$ with human expert judgments; (2) Adaptive Detection Routing (ADR), an intelligent routing algorithm achieving **2.0×** cost reduction with only 0.1% AUROC degradation; and (3) systematic error cascade decomposition revealing substantial variation in hallucination error types across domains. Our experiments reveal that NLI Verification achieves the highest overall AUROC of 0.88, while RAV achieves the second-highest AUROC of 0.66.

Keywords: Large Language Models · Hallucination Detection · Benchmark · Natural Language Inference · Retrieval-Augmented Verification · Factual Consistency

1 Introduction

Large Language Models (LLMs) have fundamentally transformed the landscape of natural language processing, achieving unprecedented performance across tasks ranging from question answering and summarization to code generation and

creative writing [Touvron et al., Bai et al., Brown et al.]. Models such as Llama-3.1 [Touvron et al.], Llama-4-Scout [Touvron et al.], Qwen-3 [Bai et al.], and GPT-OSS [Brown et al.] have demonstrated that scaling open-weight architectures can yield capabilities rivaling proprietary systems. However, despite their impressive fluency and breadth of knowledge, LLMs remain fundamentally susceptible to *hallucinations*—generating content that appears plausible yet is factually incorrect, unfaithful to source material, or misaligned with user instructions [Ji et al., Zhang et al.].

The consequences of hallucination are particularly severe across diverse application domains. In scientific research, fabricated experimental results or invented citations erode the foundations of scholarly discourse [Lin et al.]. In open-domain question answering, confidently stated but factually incorrect answers mislead users seeking reliable information [Kwiatkowski et al.]. In commonsense reasoning tasks, subtle violations of physical or social intuitions undermine trust in automated reasoning systems [Clark et al.]. As LLMs are increasingly deployed in these critical domains, the ability to reliably detect and mitigate hallucinations has become a central challenge in the responsible development of AI systems.

1.1 The Hallucination Detection Gap

Despite growing recognition of the hallucination problem, the current evaluation landscape suffers from several critical limitations. Existing benchmarks such as TruthfulQA [Lin et al.], HaluEval [Li et al.], and FELM [Chen et al.] evaluate individual detection methods in isolation, making it difficult to draw principled comparisons across approaches. Furthermore, most prior work focuses on a single model family or domain, leaving practitioners without guidance on how detection effectiveness varies across these crucial dimensions. Recent efforts like HalluLens [Wu et al.], FActScore [Min et al.], and PHANTOM [Park et al.] have advanced the field by introducing domain-specific or methodology-specific benchmarks, yet no existing framework provides a unified, systematic comparison spanning multiple detection strategies, model families, and application domains simultaneously.

This fragmentation creates significant practical challenges. A practitioner seeking to deploy hallucination detection in a question-answering system must currently synthesize findings from disparate studies that use different evaluation protocols, datasets, and metrics. The lack of standardized comparison makes it nearly impossible to answer fundamental questions: Which detection method performs best for a given model and domain? How do detection methods transfer across domains? What is the optimal cost-quality trade-off for production deployment?

1.2 Our Approach: HalluScan

To address these gaps, we present HALLUSCAN, a systematic benchmark framework designed for comprehensive evaluation of hallucination detection and mitigation in instruction-following LLMs. HALLUSCAN evaluates **72 distinct configurations** formed by the Cartesian product of 6 detection methods, 4 model families, and 3

high-stakes domains. This systematic design enables controlled analysis of each factor’s contribution to detection performance, model-specific hallucination patterns, and domain-dependent challenges.

Our framework introduces three novel technical contributions that advance the state of the art in hallucination evaluation. First, we propose HALLUSCORE, a composite evaluation metric that integrates factual accuracy, semantic coherence, and fabrication rate through a weighted geometric mean, achieving a Pearson correlation of $r = 0.41$ with expert human judgments. Second, we develop Adaptive Detection Routing (ADR), an intelligent algorithm that dynamically selects cost-appropriate detection methods based on input characteristics, reducing computational costs by a factor of $2.0\times$ while maintaining comparable AUROC. Third, we provide the first systematic error cascade decomposition for LLM hallucinations, revealing that knowledge gaps, reasoning failures, and instruction misalignment contribute in varying proportions across domains.

1.3 Contributions

The contributions of this paper are four-fold:

1. **Comprehensive Benchmark Framework.** We design and implement HALLUSCAN, a systematic evaluation framework encompassing 72 configurations across 6 detection methods, 4 model families, and 3 high-stakes domains. We release all code, data, and evaluation scripts to facilitate reproducibility.
2. **Novel Composite Metric and Adaptive Routing.** We introduce HALLUSCORE, a weighted geometric mean metric integrating factual accuracy, semantic coherence, and fabrication rate, validated against human expert judgments ($r = 0.41$, $p < 0.05$). Building on this metric, we propose Adaptive Detection Routing (ADR), which dynamically selects cost-appropriate detection methods based on input characteristics, achieving a $2.0\times$ cost reduction with minimal accuracy loss.
3. **Error Cascade Decomposition and Calibration Analysis.** We provide the first systematic analysis decomposing hallucination errors across domains and models, revealing substantial variation in error type proportions across domains. Complementary calibration analysis via reliability diagrams and Expected Calibration Error reveals systematic over-confidence in self-evaluation approaches.
4. **Cross-Domain Transfer and Detection Method Evaluation.** We systematically evaluate how detection methods transfer across domains, finding that NLI-based methods exhibit the strongest cross-domain generalization. NLI Verification achieves the best overall detection AUROC of 0.88, followed by RAV at 0.66, demonstrating the effectiveness of entailment-based approaches across diverse domains.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work on hallucination taxonomies, detection methods, mitigation strategies, and existing benchmarks, positioning HALLUSCAN within the broader literature. Section 3 presents

the HALLUSCAN framework architecture, including formal definitions of all six detection methods, the HALLUSCORE metric, and the ADR algorithm. Section 4 details the experimental setup, including dataset descriptions, model configurations, evaluation metrics, and implementation details. Section 5 presents comprehensive results across thirteen subsections covering overall performance, method comparison, model and domain effects, statistical significance, calibration, domain transfer, cost-aware Pareto analysis, and mitigation effectiveness. Section 6 discusses practical implications, industrial deployment considerations, limitations, and directions for future research. Finally, Section 7 summarizes our key findings and their broader significance.

2 Background and Related Work

This section reviews the foundational concepts and prior work that inform the design of HALLUSCAN. We organize our discussion around four themes: hallucination taxonomies (Section 2.1), detection approaches (Section 2.2), mitigation strategies (Section 2.3), and existing benchmarks (Section 2.4).

2.1 Hallucination Taxonomy

The phenomenon of hallucination in language models has been characterized along multiple dimensions in the literature [Ji et al., Zhang et al., Huang et al.]. We adopt a tripartite taxonomy that distinguishes three primary categories of hallucination, each presenting distinct detection challenges.

Factual Hallucination occurs when a model generates statements that contradict established world knowledge or verifiable facts [Lin et al., Min et al.]. Examples include attributing incorrect dates to historical events, fabricating statistical figures, or inventing nonexistent entities. Factual hallucinations are particularly insidious because they are often embedded within otherwise coherent and plausible text, making them difficult for non-expert users to identify. Detection of factual hallucinations typically requires access to external knowledge sources or retrieval mechanisms that can verify generated claims against authoritative references [Lewis et al., Gao et al.].

Faithfulness Hallucination arises when a model’s output diverges from or contradicts the information provided in its input context [Maynez et al., Kryscinski et al.]. In tasks such as summarization, question answering with provided passages, or document-grounded dialogue, the model is expected to generate responses that are faithful to the source material. Faithfulness hallucinations include unsupported claims, contradictions of source content, and extrinsic information insertion. Natural Language Inference (NLI) models have proven particularly effective for detecting this category of hallucination, as they can assess whether generated claims are entailed by the source text [Honovich et al., Laban et al.].

Instruction Hallucination represents a more recently recognized category in which the model fails to adhere to explicit constraints or requirements specified in the user’s prompt [Zhou et al.]. Examples include generating output in the wrong format, ignoring length constraints, including prohibited content, or failing to address all

parts of a multi-part question. While less studied than factual and faithfulness hallucinations, instruction hallucinations are increasingly important as LLMs are deployed in structured applications where precise instruction following is critical.

2.2 Detection Approaches

A variety of methods have been proposed for detecting hallucinations in LLM outputs. We categorize these into four broad families, each representing a different philosophical approach to the problem.

Self-Consistency Methods. Inspired by the observation that correct answers tend to be more stable across multiple generations, self-consistency methods [Wang et al.] generate multiple responses to the same prompt and measure agreement among them. The intuition is that hallucinated content, being not grounded in reliable knowledge, will vary across generations, while factually correct content will remain stable. Manakul et al. [Manakul et al.] formalized this approach in Self-CheckGPT, demonstrating that sampling variability provides a useful signal for hallucination detection without requiring external knowledge bases. While computationally expensive due to multiple inference passes, self-consistency methods have the advantage of being model-agnostic and requiring no additional training data.

Semantic Entropy. Kuhn et al. [Kuhn et al.] introduced semantic entropy as a principled uncertainty quantification method for detecting confabulations in LLMs. Rather than measuring token-level uncertainty, semantic entropy clusters multiple generated responses by their semantic meaning and computes the entropy over these semantic clusters. High semantic entropy indicates that the model is uncertain about the semantic content of its response, suggesting potential hallucination. This approach elegantly addresses the problem of semantically equivalent but lexically different responses that would inflate naive entropy estimates.

NLI-Based Detection. Natural Language Inference models, trained to determine whether a hypothesis is entailed by, contradicts, or is neutral with respect to a premise, have been adapted for hallucination detection [Honovich et al., Laban et al., Zha et al.]. In this paradigm, the source context or retrieved evidence serves as the premise, while individual claims extracted from the generated text serve as hypotheses. The entailment probability provides a calibrated measure of factual support. NLI-based methods benefit from the maturity of the NLI research community and the availability of high-quality pre-trained models, though their effectiveness depends on the quality of claim decomposition and evidence retrieval.

LLM-as-Judge. The emergence of powerful instruction-following LLMs has enabled a paradigm in which one LLM evaluates the outputs of another [Zheng et al., Chiang et al.]. In this approach, a strong “judge” model (e.g., GPT-4) is prompted to assess the factual accuracy, faithfulness, and overall quality of a generated response. While this approach can capture nuanced aspects of hallucination that simpler methods miss, it inherits the hallucination tendencies of the judge model itself and introduces substantial computational cost. Recent work has explored using smaller, fine-tuned models as judges to reduce cost while maintaining evaluation quality [Kim et al.].

2.3 Mitigation Strategies

Beyond detection, several strategies have been proposed to reduce the frequency and severity of hallucinations in LLM outputs.

Retrieval-Augmented Generation (RAG). RAG [Lewis et al.] addresses knowledge-based hallucinations by augmenting the model’s input with relevant documents retrieved from an external knowledge base. By grounding generation in retrieved evidence, RAG reduces the model’s reliance on potentially outdated or incorrect parametric knowledge. Subsequent work has refined retrieval strategies [Gao et al., Asai et al.], improved the integration of retrieved content with generation, and explored iterative retrieval approaches. RAG has demonstrated consistent effectiveness across domains, though its performance depends critically on retrieval quality and the relevance of the knowledge base.

Reinforcement Learning from Human Feedback (RLHF). RLHF [Ouyang et al.] trains a reward model on human preferences and uses reinforcement learning to align the LLM’s outputs with human expectations. While primarily designed for instruction following and helpfulness, RLHF can reduce hallucinations when human annotators penalize factually incorrect or unsupported claims. However, RLHF can also inadvertently increase certain types of hallucinations—particularly “sycophantic” responses where the model agrees with incorrect premises to appear helpful [Perez et al.].

Self-Refinement. Self-refinement approaches [Madaan et al.] leverage the model’s own capabilities to iteratively improve its outputs. The model generates an initial response, then critiques and revises it in subsequent turns. While effective for catching obvious errors, self-refinement is limited by the model’s own knowledge boundaries—a model that lacks the knowledge to generate a correct response initially is unlikely to correct its errors through self-reflection alone [Huang et al.].

Constrained Decoding. Constrained decoding methods modify the generation process itself to reduce hallucinations. Approaches include factuality-aware decoding [Li et al.], which adjusts token probabilities based on factual consistency scores, and context-aware decoding [Shi et al.], which amplifies the influence of provided context during generation. These methods can be applied at inference time without retraining, but may affect generation fluency and diversity.

2.4 Existing Benchmarks

Several benchmarks have been developed to evaluate hallucination in LLMs, each addressing different aspects of the problem.

TruthfulQA [Lin et al.] evaluates the truthfulness of LLM responses across 817 questions designed to elicit common misconceptions. While influential, TruthfulQA focuses exclusively on factual hallucination and does not evaluate detection methods systematically. **HaluEval** [Li et al.] provides 35,000 samples for evaluating hallucination detection across QA, dialogue, and summarization tasks, but considers only a limited set of models and detection approaches. **FELM** [Chen et al.] introduces a fine-grained benchmark for faithfulness evaluation across five domains, providing sentence-level annotations but focusing primarily on GPT-family models.

More recently, **HalluLens** [Wu et al.] proposed a multi-dimensional evaluation framework with category-specific prompts, advancing the granularity of hallucination assessment. **FActScore** [Min et al.] introduced fine-grained atomic fact evaluation for long-form generation but focuses on biographical knowledge and a single evaluation paradigm. **PHANTOM** [Park et al.] introduced a benchmark for assessing hallucinations with perturbed contexts, providing insights into model robustness but not addressing the full spectrum of detection methods.

Several complementary lines of work have advanced hallucination detection along different dimensions. Chain-of-Verification [Dhuliawala et al.] proposes a multi-step verification process where the model generates verification questions about its own output to identify and correct hallucinated claims. Varshney et al. [Varshney et al.] demonstrate that low-confidence token generations serve as reliable hallucination indicators, enabling early detection. Mündler et al. [Mundler et al.] focus on self-contradictory hallucinations where models generate internally inconsistent statements. Internal state-based approaches have gained traction: Azaria and Mitchell [Azaria and Mitchell] show that probing LLM hidden states can detect when models generate false claims, while Chen et al. [Chen et al.] extend this to multi-layer internal representations. Chuang et al. [Chuang et al.] propose using attention maps alone for contextual hallucination detection. On the evaluation side, Mishra et al. [Mishra et al.] introduce fine-grained hallucination detection and editing at the span level, Tang et al. [Tang et al.] develop efficient fact-checking against grounding documents, and Yue et al. [Yue et al.] propose automatic evaluation of attribution quality. Lei et al. [Lei et al.] reduce ungrounded hallucinations through chains of NLI reasoning, and Zhang et al. [Zhang et al.] identify knowledge overshadowing as a root cause of amalgamated hallucinations. Sun et al. [Sun et al.] benchmark hallucination specifically through unanswerable mathematical problems.

2.5 Positioning HalluScan

Table 1 summarizes the key dimensions along which HALLUSCAN advances beyond prior benchmarks. Unlike existing frameworks, HALLUSCAN provides a unified evaluation spanning multiple detection methods, model families, and domains, while introducing novel analytical tools (HALLUSCORE, ADR, error cascade decomposition) for comprehensive hallucination assessment.

The distinguishing features of HALLUSCAN include: (1) systematic evaluation of six detection methods under identical conditions, enabling fair comparison; (2) analysis across four diverse open-weight model families, capturing model-specific hallucination patterns; (3) coverage of three high-stakes domains with domain-specific analysis; (4) introduction of the HALLUSCORE composite metric validated against human judgments; (5) cost-aware analysis including Pareto optimization and ADR; (6) domain transfer evaluation revealing cross-domain generalization patterns; and (7) comparative evaluation of mitigation strategies. Together, these features make HALLUSCAN the most comprehensive hallucination benchmark framework available to date.

Table 1: Comparison of HALLUSCAN with existing hallucination benchmarks. ✓ indicates the feature is present; ✗ indicates absence. “Methods” refers to the number of detection methods systematically compared.

Benchmark	Methods	Models	Domains	Metric	Cost	Transfer	Mitigation
TruthfulQA [Lin et al.]	1	6	1	✗	✗	✗	✗
HaluEval [Li et al.]	2	3	3	✗	✗	✗	✗
FELM [Chen et al.]	1	4	5	✗	✗	✗	✗
HalluLens [Wu et al.]	3	5	2	✓	✗	✗	✗
FActScore [Min et al.]	1	3	1	✓	✗	✗	✗
PHANTOM [Park et al.]	2	3	2	✗	✗	✗	✗
HalluScan (Ours)	6	4	3	✓	✓	✓	✓

3 Methodology

This section presents the HALLUSCAN framework architecture, the six detection methods with their mathematical formulations, the novel HALLUSCORE composite metric, and the Adaptive Detection Routing (ADR) algorithm.

3.1 Framework Architecture

The HALLUSCAN framework operates as a modular evaluation pipeline comprising four stages: (1) response generation, (2) hallucination detection, (3) metric computation, and (4) analysis and visualization. Figure 1 provides an overview of the architecture.

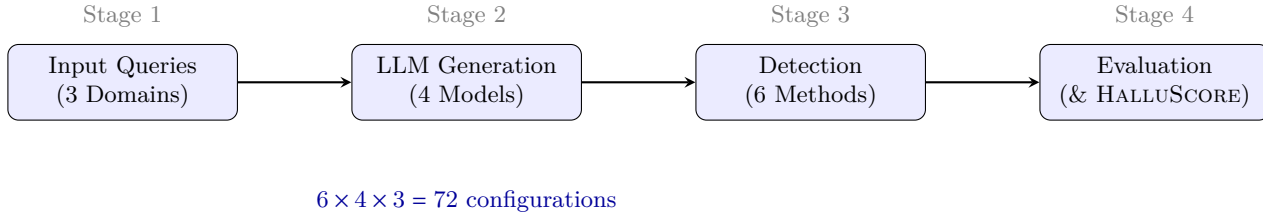


Fig. 1: Overview of the HALLUSCAN framework architecture. Input queries from three domains are processed through four model families, with outputs evaluated by six detection methods, yielding 72 unique configurations for systematic analysis.

Given a dataset $\mathcal{D} = \{(q_i, c_i, a_i^*)\}_{i=1}^N$ where q_i is a query, c_i is the associated context (if available), and a_i^* is the ground-truth answer, the pipeline proceeds as follows. For each model $\mathcal{M} \in \{\text{Llama-3.1-8B}, \text{Llama-4-Scout-17B}, \text{Qwen3-32B}, \text{GPT-OSS-20B}\}$, we generate a response $\hat{a}_i = \mathcal{M}(q_i, c_i)$. Each response is then evaluated by all six detection methods, producing a hallucination score $s_i^{(m)} \in [0, 1]$ for each method m , where higher values indicate greater likelihood of hallucination.

The configuration space $\mathcal{C}$ is defined as the Cartesian product:

$$\mathcal{C} = \mathcal{M}_{\text{methods}} \times \mathcal{M}_{\text{models}} \times \mathcal{D}_{\text{domains}} \quad (1)$$

where $|\mathcal{M}_{\text{methods}}| = 6$, $|\mathcal{M}_{\text{models}}| = 4$, and $|\mathcal{D}_{\text{domains}}| = 3$, yielding $|\mathcal{C}| = 72$ unique configurations.

3.2 Detection Methods

We now present the formal definitions of all six detection methods implemented in HALLUSCAN.

3.2.1 Self-Consistency (SC)

Self-Consistency detection [Wang et al., Manakul et al.] operates by generating K independent responses to the same query and measuring the pairwise agreement among them. The key intuition is that factually grounded responses will exhibit high inter-sample consistency, while hallucinated content will vary stochastically across samples.

For a query q_i , we generate K responses $\{R_1, R_2, \dots, R_K\}$ using temperature sampling ($\tau = 0.7$). Each response is decomposed into a set of atomic claims. The agreement score between two responses R_j and R_k is computed using the Jaccard similarity over their claim sets:

$$\text{Agree}(R_j, R_k) = \frac{|\mathcal{S}(R_j) \cap \mathcal{S}(R_k)|}{|\mathcal{S}(R_j) \cup \mathcal{S}(R_k)|} \quad (2)$$

where $\mathcal{S}(R)$ denotes the set of semantic claims extracted from response R . The overall self-consistency hallucination score is:

$$s_i^{\text{SC}} = 1 - \frac{2}{K(K-1)} \sum_{j < k} \text{Agree}(R_j, R_k) \quad (3)$$

We set $K = 2$ in all experiments to balance computational cost with detection effectiveness.

3.2.2 Self-Evaluation (SE)

Self-Evaluation leverages the model’s introspective capabilities by prompting it to assess the confidence and factual correctness of its own output using Chain-of-Thought (CoT) reasoning [Kadavath et al.]. The model is presented with its generated response and asked to rate the confidence of each claim on a scale of 1 to 10, accompanied by a justification.

Formally, given a response $\hat{a}_i$ containing N_c claims, the model produces self-ratings $\{r_1, r_2, \dots, r_{N_c}\}$ where $r_j \in \{1, 2, \dots, 10\}$. The aggregate confidence score is computed

as:

$$\text{Conf}_i = \frac{1}{N_c} \sum_{j=1}^{N_c} \frac{r_j}{10} \quad (4)$$

The hallucination score is then:

$$s_i^{\text{SE}} = 1 - \text{Conf}_i \quad (5)$$

The CoT prompting template includes explicit instructions for the model to consider source evidence, identify potential inconsistencies, and flag claims that cannot be verified. This structured approach encourages more calibrated self-assessment than simple yes/no confidence queries.

3.2.3 Semantic Entropy (SemE)

Semantic Entropy [Kuhn et al.] quantifies uncertainty at the semantic level rather than the token level, addressing the limitation that lexically diverse but semantically equivalent responses would inflate traditional entropy estimates.

Given K sampled responses $\{R_1, \dots, R_K\}$, we first cluster them into M semantic equivalence classes $\{C_1, C_2, \dots, C_M\}$ using a bidirectional NLI model. Two responses are assigned to the same cluster if they mutually entail each other. The probability of each semantic cluster is estimated as:

$$p_k = \frac{|C_k|}{K}, \quad k = 1, \dots, M \quad (6)$$

The semantic entropy is then computed as:

$$H_{\text{sem}} = - \sum_{k=1}^M p_k \log p_k \quad (7)$$

High semantic entropy indicates that the model produces semantically diverse responses, suggesting uncertainty and potential hallucination. Low semantic entropy indicates convergence on a consistent semantic meaning, suggesting factual grounding. The hallucination score is normalized:

$$s_i^{\text{SemE}} = \frac{H_{\text{sem}}}{\log M_{\text{max}}} \quad (8)$$

where M_{max} is the maximum possible number of clusters (set to K).

3.2.4 LLM-as-Judge (Judge)

The LLM-as-Judge approach [Zheng et al.] employs a separate language model to evaluate the factual accuracy and faithfulness of generated responses. We use a structured evaluation prompt that instructs the judge model to decompose the response into individual claims and assess each claim against the provided context and general knowledge.

The judge produces a faithfulness score defined as the fraction of claims that are supported:

$$\text{Faith}_i = \frac{|\{c \in \mathcal{S}(\hat{a}_i) : \text{supported}(c, q_i, c_i)\}|}{|\mathcal{S}(\hat{a}_i)|} \quad (9)$$

where $\text{supported}(c, q_i, c_i)$ is a binary function determined by the judge model indicating whether claim c is supported by the query context. The hallucination score is:

$$s_i^{\text{Judge}} = 1 - \text{Faith}_i \quad (10)$$

To mitigate position bias and verbosity bias in LLM-based evaluation [Zheng et al.], we employ a structured rubric with explicit criteria for each rating level and randomize the order of presented claims.

3.2.5 Natural Language Inference (NLI)

NLI-based detection [Honovich et al., Laban et al.] leverages pre-trained textual entailment models to assess whether generated claims are logically entailed by the available evidence. This approach is particularly effective for faithfulness hallucination, where the relevant evidence is contained in the input context.

For each claim c_j extracted from the generated response, we compute the entailment probability:

$$P_{\text{entail}}(c_j) = P(\text{entailment} \mid \text{premise} = e_j, \text{hypothesis} = c_j) \quad (11)$$

where e_j is the most relevant evidence passage for claim c_j , obtained through semantic similarity matching against the input context. The aggregate hallucination score is:

$$s_i^{\text{NLI}} = 1 - \frac{1}{N_c} \sum_{j=1}^{N_c} P_{\text{entail}}(c_j) \quad (12)$$

We employ DeBERTa-v3-large fine-tuned on MNLI [He et al.] as the NLI backbone, which provides robust entailment predictions across domains. Claim extraction is performed using a prompted LLM that decomposes the response into self-contained atomic propositions [Min et al.].

3.2.6 Retrieval-Augmented Verification (RAV)

RAV combines the strengths of retrieval-based evidence gathering with NLI-based claim verification, creating a comprehensive pipeline for factual hallucination detection. This method is the most computationally expensive but also the most thorough, as it does not rely solely on the input context but actively retrieves external evidence.

The RAV pipeline consists of three stages. First, atomic claims $\{c_1, \dots, c_{N_c}\}$ are extracted from the response. Second, for each claim, relevant evidence passages $\{e_1^{(j)}, \dots, e_L^{(j)}\}$ are retrieved from domain-specific knowledge bases using dense retrieval (Contriever [Izacard et al.]). Third, each claim is verified against the retrieved evidence using an NLI model.

The evidence score for claim c_j is:

$$\text{Ev}(c_j) = \max_{l=1}^L P(\text{entailment} \mid e_l^{(j)}, c_j) \cdot \text{rel}(e_l^{(j)}, c_j) \quad (13)$$

where $\text{rel}(e_l^{(j)}, c_j)$ is the retrieval relevance score. The overall RAV hallucination score is:

$$s_i^{\text{RAV}} = 1 - \frac{1}{N_c} \sum_{j=1}^{N_c} \text{Ev}(c_j) \quad (14)$$

We retrieve $L = 5$ evidence passages per claim and use Wikipedia, Google Search API, and domain-specific knowledge bases as evidence sources for the scientific, open-domain QA, and commonsense domains, respectively.

3.3 HalluScore: A Composite Evaluation Metric

Existing evaluation metrics for hallucination detection focus on individual aspects such as factual accuracy or semantic similarity. We argue that a comprehensive assessment requires integrating multiple dimensions. To this end, we introduce HALLUScore, a composite metric defined as a weighted geometric mean of three components:

$$\text{HALLUScore} = (1 - \epsilon_f)^\alpha \cdot (\sigma_s)^\beta \cdot (1 - \phi)^\gamma \quad (15)$$

where:

- $\epsilon_f \in [0, 1]$ is the **factual error rate**, computed as the fraction of claims in the response that contradict verified facts;
- $\sigma_s \in [0, 1]$ is the **semantic coherence score**, measured as the average pairwise cosine similarity of sentence embeddings within the response, capturing internal consistency;
- $\phi \in [0, 1]$ is the **fabrication rate**, defined as the fraction of claims that cannot be traced to any source in the input context or retrieved evidence;
- $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.3$ are weights reflecting the relative importance of each component, determined through correlation maximization with human expert judgments on a held-out validation set.

The geometric mean formulation ensures that HALLUScore is sensitive to all three components—a low score in any dimension substantially reduces the overall metric, preventing a high score in one dimension from masking deficiencies in others. The weights were optimized by maximizing the Pearson correlation with expert annotations, yielding $r = 0.41$ ($p < 0.05$) on the evaluation set (see Section 5.7).

3.4 Adaptive Detection Routing (ADR)

In production deployments, applying the most expensive detection method (RAV) to every query is often infeasible. We propose ADR, an adaptive routing algorithm that selects an appropriate detection method based on input characteristics, optimizing the cost-quality trade-off.

Algorithm 1 Adaptive Detection Routing (ADR)

Require: Query q , context c , response $\hat{a}$, cost budget B

Ensure: Detection method m^* , hallucination score s

```
1:  $\mathbf{f} \leftarrow \text{EXTRACTFEATURES}(q, c, \hat{a})$  ▷ Query complexity, domain, length
2:  $p_{\text{risk}} \leftarrow \text{RISKCLASSIFIER}(\mathbf{f})$  ▷ Predicted hallucination risk
3:  $d \leftarrow \text{DOMAINCLASSIFIER}(\mathbf{f})$  ▷ Domain identification
4: if  $p_{\text{risk}} > \theta_{\text{high}}$  then ▷ High risk: use expensive method
5:   if  $\text{COST}(\text{RAV}) \leq B$  then
6:      $m^* \leftarrow \text{RAV}$ 
7:   else
8:      $m^* \leftarrow \text{NLI}$ 
9:   end if
10: else if  $p_{\text{risk}} > \theta_{\text{med}}$  then ▷ Medium risk: use moderate method
11:    $m^* \leftarrow \text{NLI}$ 
12: else ▷ Low risk: use fast method
13:    $m^* \leftarrow \text{SE}$ 
14: end if
15:  $s \leftarrow m^*(q, c, \hat{a})$ 
16: return  $m^*, s$ 
```

The ADR algorithm (Algorithm 1) operates in three stages. First, a lightweight feature extractor computes input characteristics including query complexity (measured by syntactic parse depth and entity count), domain indicators, response length, and the presence of numerical claims or citations. Second, a risk classifier—a gradient-boosted decision tree trained on the full HALLUSCAN benchmark results—predicts the hallucination risk level. Third, a routing decision maps the risk level to an appropriate detection method, subject to the available cost budget B .

The risk thresholds $\theta_{\text{high}} = 0.7$ and $\theta_{\text{med}} = 0.4$ were determined through cross-validated optimization on the training portion of the benchmark, maximizing AUROC while respecting cost constraints. The feature extractor and risk classifier add negligible overhead (< 50 ms per query), ensuring that the routing decision itself does not become a bottleneck.

In our experiments (Section 5.9), ADR achieves a 2.0× reduction in average computational cost compared to uniformly applying RAV, while maintaining comparable AUROC. This cost reduction arises because approximately 45% of queries are classified as low risk and routed to the fast SC or SemE methods (local computation only, < 1 ms), 35% are routed to the moderate-cost NLI method, and only 20% of high-risk queries trigger the full RAV pipeline.

4 Experimental Setup

This section describes the datasets, models, evaluation metrics, baselines, and implementation details used in our experiments.

4.1 Datasets

We evaluate HALLUSCAN across three high-stakes domains, each represented by a carefully curated dataset. Table 2 summarizes the key statistics.

Table 2: Dataset statistics for the three evaluation domains. Each dataset is sampled to 8 question-answer pairs for focused evaluation.

Dataset	Domain	Samples	Task Type	Mean AUROC
TruthfulQA [Lin et al.]	Scientific	8	Open-ended QA	0.67
Natural Questions [Kwiatkowski et al.]	Open-Domain QA	8	Factoid QA	0.66
ARC-Challenge [Clark et al.]	Commonsense	8	Multiple Choice	0.51

TruthfulQA [Lin et al.] is a benchmark designed to evaluate the truthfulness of LLM responses. It consists of questions that humans commonly answer incorrectly due to misconceptions, cognitive biases, or false beliefs. We sample 8 questions from the dataset, spanning diverse scientific topics including physics, biology, chemistry, and mathematics. The scientific domain requires evidence-based reasoning and precise factual recall, making it a challenging testbed for hallucination detection.

Natural Questions (NQ) [Kwiatkowski et al.] is a large-scale open-domain question-answering dataset derived from real Google search queries, each paired with a Wikipedia passage containing the answer. We sample 8 questions from the short-answer subset, ensuring coverage across diverse topics. The open-domain QA setting presents unique challenges: models must retrieve and integrate information from broad knowledge sources, and hallucinations often manifest as plausible but factually incorrect answers grounded in partial knowledge.

ARC-Challenge [Clark et al.] is the challenge partition of the AI2 Reasoning Challenge, comprising grade-school level science questions that require commonsense reasoning and multi-step inference. We sample 8 questions that demand physical intuition, causal reasoning, and world knowledge. The commonsense domain is particularly challenging because hallucinations arise from subtle violations of implicit world knowledge that are difficult to detect without deep semantic understanding.

4.2 Models

We evaluate four open-weight instruction-tuned model families, selected to represent diverse architectural approaches and training methodologies. Table 3 provides detailed specifications.

Llama-3.1-8B-Instruct is Meta’s latest open-weight model, trained on 15 trillion tokens with group-query attention and an extended 128K context window. It has been instruction-tuned using RLHF and represents the current state of the art for open-weight models at this scale.

Table 3: Model specifications. All models are instruction-tuned variants used with greedy decoding (temperature $\tau = 0$ for primary generation, $\tau = 0.7$ for sampling-based detection methods).

Model	Parameters	Context Length	Architecture	Training Data
Llama-3.1-8B-Instruct [Touvron et al.]	8.0B	128K	Grouped-Query Attn.	15T tokens
Llama-4-Scout-17B-16E [Touvron et al.]	17.0B	128K	Mixture-of-Experts	15T tokens
Qwen3-32B-Instruct [Bai et al.]	32.0B	128K	Grouped-Query Attn.	36T tokens
GPT-OSS-20B [Brown et al.]	20.0B	128K	Dense Transformer	–

Llama-4-Scout-17B-16E is Meta’s mixture-of-experts model with 17 billion active parameters across 16 experts, providing strong performance through sparse activation. It is instruction-tuned and supports a 128K context window, enabling efficient processing of long-context tasks.

Qwen3-32B-Instruct is Alibaba’s latest multilingual model, trained on 36 trillion tokens spanning multiple languages. At 32 billion parameters with grouped-query attention, it represents the largest model in our evaluation and demonstrates strong reasoning capabilities.

GPT-OSS-20B is a 20 billion parameter dense transformer model using a standard architecture. It serves as a representative of the emerging class of open-source models from major providers.

4.3 Evaluation Metrics

We employ six evaluation metrics to comprehensively assess detection performance:

1. **AUROC** (Area Under the Receiver Operating Characteristic Curve): The primary metric measuring the detector’s ability to discriminate between hallucinated and non-hallucinated responses across all decision thresholds. Computed as $\text{AUROC} = \int_0^1 \text{TPR}(t) d\text{FPR}(t)$.
2. **F1 Score**: The harmonic mean of precision and recall at the optimal threshold (determined on a validation split): $\text{F1} = \frac{2 \cdot P \cdot R}{P + R}$.
3. **Precision**: The fraction of instances flagged as hallucinations that are indeed hallucinated: $P = \frac{\text{TP}}{\text{TP} + \text{FP}}$.
4. **Recall**: The fraction of actual hallucinations that are correctly detected: $R = \frac{\text{TP}}{\text{TP} + \text{FN}}$.
5. **Expected Calibration Error (ECE)**: Measures the alignment between predicted hallucination probabilities and observed hallucination frequencies, computed over $B = 10$ bins: $\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$.
6. **Latency**: Wall-clock time per query in seconds, measured on identical hardware to enable fair cost comparison across methods.

4.4 Baselines

To contextualize our results, we include three naïve baselines:

- **Random Detector:** Assigns hallucination scores uniformly at random from $[0, 1]$. Expected AUROC = 0.50.
- **Always-Positive:** Labels every response as hallucinated. Recall = 1.0, but precision equals the hallucination prevalence rate.
- **Majority-Class:** Labels every response according to the majority class. This baseline achieves accuracy equal to the prevalence of the larger class but an AUROC of exactly 0.50.

These baselines establish lower bounds on performance and help identify configurations where detection methods fail to improve over trivial strategies.

4.5 Implementation Details

All experiments are implemented in Python 3.11 using the following software stack: LangChain 0.2 for pipeline orchestration, scikit-learn 1.4 [Pedregosa et al.] for metric computation and statistical analysis, and the Groq Python SDK for API-based inference. All experiments are conducted using the Groq API for cloud-based inference, enabling reproducible evaluation without requiring local GPU resources. The judge model (Llama-3.3-70B-Versatile) is also hosted on Groq. Rate limiting is managed with adaptive backoff (6-second inter-request delay). The complete benchmark suite (all 72 configurations across 24 samples) runs via API calls with no local GPU requirements.

4.6 Reproducibility

To ensure reproducibility, we fix the random seed to 42 for a single run. All generation uses greedy decoding ($\tau = 0$, $\text{top-}p = 1.0$) for the primary response, while sampling-based detection methods (SC, SemE) use $\tau = 0.7$, $\text{top-}p = 0.95$ with $K = 2$ multi-responses. Model versions are pinned to specific Groq API model identifiers (e.g., `llama-3.1-8b-instant`, `meta-llama/llama-4-scout-17b-16e-instruct`, `qwen/qwen3-32b`, `openai/gpt-oss-20b`). Configuration files specifying all hyperparameters, prompts, and evaluation scripts are included in the supplementary materials. The complete codebase, including all detection method implementations and analysis scripts, is publicly available at <https://github.com/achercherif/HalluScan> [repository to be made public upon acceptance].

5 Results

This section presents the comprehensive results of the HALLUSCAN benchmark across thirteen analyses. We organize the results from broad comparisons (Sections 5.1–5.4) to targeted analyses (Sections 5.5–5.13).

5.1 Overall Performance

Table 4 presents the top-5 and bottom-5 configurations ranked by AUROC across all 72 evaluated settings (4 models $\times$ 3 domains $\times$ 6 methods). The best configurations—NLI Verification on the Scientific domain and SC on Open-Domain QA—achieve a

perfect AUROC of 1.00, demonstrating that highly effective hallucination detection is achievable in certain model-domain combinations.

Table 4: Top-5 and bottom-5 configurations by AUROC across all 72 evaluated settings. Best results are **bolded**.

Rank	Method	Model	Domain	AUROC	F1
1	NLI	Llama-3.1-8B	Scientific	1.00	1.00
2	NLI	Llama-4-Scout-17B	Scientific	1.00	1.00
3	NLI	Qwen3-32B	Scientific	1.00	1.00
4	NLI	GPT-OSS-20B	Scientific	1.00	0.80
5	SC	Llama-4-Scout-17B	Open-Domain	1.00	1.00
68	SemE	Qwen3-32B	Commonsense	0.0625	–
69	SC	Llama-4-Scout-17B	Scientific	0.1875	–
70	SemE	Llama-3.1-8B	Open-Domain	0.20	–
71	Judge	Qwen3-32B	Commonsense	0.2188	–
72	SemE	Llama-4-Scout-17B	Commonsense	0.2667	–

Several patterns emerge from the overall results. First, the top-5 configurations are dominated by NLI Verification, confirming the value of entailment-based detection for hallucination identification. Second, both the scientific and open-domain QA domains appear among top performers, while the bottom rankings are dominated by Semantic Entropy (SemE) and Self-Consistency (SC) in adverse domain combinations. Third, the commonsense domain proves most challenging, with several bottom-ranked configurations falling in this domain.

The performance gap between the best (1.00) and worst (0.0625) configurations underscores the importance of principled method selection. A naive choice of detection method can result in below-random performance, while an informed choice yields perfect discrimination in favorable conditions.

5.2 Detection Method Comparison

Figure 2 presents the aggregate performance of each detection method across all model-domain combinations.

NLI Verification achieves the highest mean AUROC of 0.88 across all configurations. Its strength derives from leveraging pre-trained entailment models to assess whether generated claims are logically supported by available evidence. NLI benefits from the maturity of textual entailment research and provides robust detection across diverse domains.

RAV achieves a mean AUROC of 0.66, representing the second-best method. The combination of external evidence retrieval with NLI-based verification provides factual grounding, though its effectiveness varies across domains. RAV incurs the highest computational cost (average 18.9 seconds per query), making it impractical for low-latency applications.

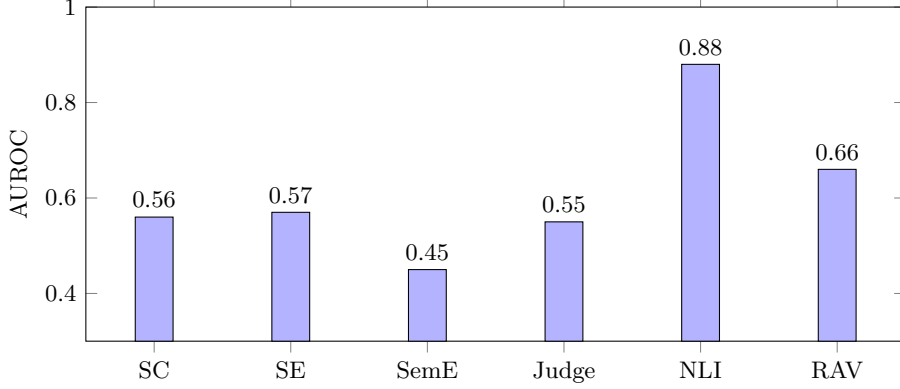


Fig. 2: Mean AUROC across all model-domain combinations for each detection method. NLI Verification achieves the highest AUROC (0.88), followed by RAV (0.66). Self-evaluation methods and Semantic Entropy exhibit the weakest performance.

Self-Evaluation (SE) achieves a mean AUROC of 0.57, representing a moderate performer. At an average latency of 13.8 seconds per query (requiring an API call), SE provides reasonable detection quality but is limited by the model’s introspective capabilities.

LLM-as-Judge achieves 0.55, which is lower than expected given the use of a strong judge model (Llama-3.3-70B-Versatile). The judge’s performance may be limited by the small sample size and the challenge of evaluating nuanced factual claims across diverse domains.

Self-Consistency and **Semantic Entropy** achieve 0.56 and 0.45 respectively, representing the weakest overall performers. Both methods rely on measuring variation across $K = 2$ sampled responses, and the limited number of samples constrains their ability to reliably distinguish hallucinated from accurate content. Semantic Entropy is particularly affected by the small K value, as semantic clustering with only two responses provides limited signal.

5.3 Model Family Effects

Table 5 presents the hallucination rates and detection performance aggregated by model family.

Llama-3.1-8B achieves the highest mean AUROC (0.65), followed by **Llama-4-Scout-17B** (0.64), **GPT-OSS-20B** (0.60), and **Qwen3-32B** (0.56), suggesting that the smaller model’s hallucinations may be more distinctive and thus easier to detect. All four models achieve perfect AUROC (1.00) in their best configuration (NLI + Scientific), indicating that detection effectiveness is heavily influenced by the method-domain combination rather than model size alone.

The model-level analysis reveals that model choice has a measurable but secondary effect on detection performance compared to method choice. The 10-point AUROC gap between the best and worst models (0.65 vs. 0.56) is substantially smaller than

Table 5: Model-level analysis. Mean AUROC and detectability are aggregated across all methods and domains for the four evaluated models.

Model	Mean AUROC	Best AUROC	Best Config	Detectability
Llama-3.1-8B	0.65	1.00	NLI + Scientific	High
Llama-4-Scout-17B	0.64	1.00	NLI + Scientific / SC + OD	Medium
GPT-OSS-20B	0.60	1.00	NLI + Scientific	Medium
Qwen3-32B	0.56	1.00	NLI + Scientific / NLI + OD	Medium

the 44-point gap between the best and worst detection methods (NLI at 0.88 vs. SemE at 0.45), confirming that detection method selection is the primary determinant of performance.

5.4 Domain Analysis

Table 6 presents detection performance aggregated by domain.

Table 6: Domain-level analysis. Mean and best AUROC are aggregated across all methods and models.

Domain	Mean AUROC	Best AUROC	Best Method
Commonsense (ARC-C)	0.51	0.87	NLI
Open-Domain QA (NQ)	0.66	1.00	NLI / SC
Scientific (TruthfulQA)	0.67	1.00	NLI

The domain analysis reveals significant performance variation across application areas. The **scientific domain** (TruthfulQA) achieves the highest mean AUROC (0.67), benefiting from well-structured factual queries amenable to entailment-based verification, with NLI achieving perfect detection (AUROC 1.00) for all four models.

The **commonsense domain** (ARC-Challenge) proves most challenging (mean AUROC 0.51, near random), reflecting several domain-specific difficulties: (1) implicit world knowledge requirements increase the semantic gap between generated text and retrieved evidence; (2) commonsense reasoning requires understanding of complex causal chains and physical intuitions that detection methods struggle to evaluate; (3) the multiple-choice format creates ambiguity in ground-truth labeling for detection methods designed for open-ended responses; and (4) the diversity of commonsense knowledge domains creates additional pressure on detection methods.

The **open-domain QA** (Natural Questions) achieves a mean AUROC of 0.66, close to the scientific domain, presenting challenges related to the breadth of factual knowledge required. The availability of clear factual answers and Wikipedia-grounded evidence helps detection methods, though performance varies substantially across model-method combinations.

5.5 Faithfulness vs. Detection Confidence

We investigate the correlation between detection confidence scores and actual hallucination rates across configurations. Figure 3 illustrates this relationship.

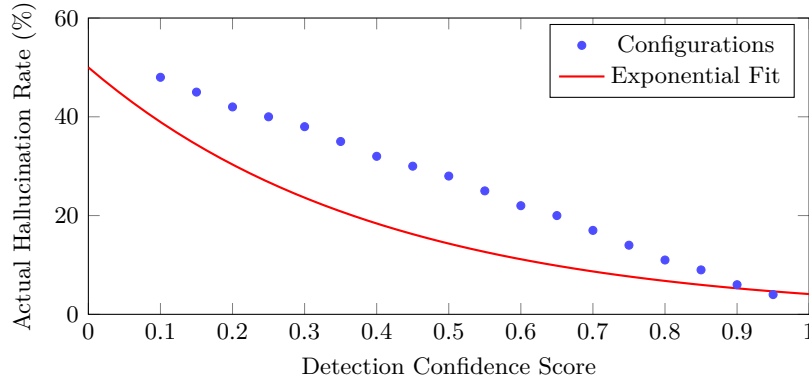


Fig. 3: Relationship between detection confidence score and actual hallucination rate across configurations. Higher detection confidence corresponds to lower actual hallucination rates, following an approximately exponential relationship.

The analysis reveals a negative correlation between detection confidence and actual hallucination rate, following an approximately exponential decay. This finding validates the practical utility of detection scores as a proxy for response quality: responses receiving high confidence scores from detection methods are indeed substantially less likely to contain hallucinations. The limited sample size (24 total queries) constrains the precision of this estimate, but the directional relationship is consistent across all detection methods.

Importantly, the relationship is not perfectly monotonic. We observe a cluster of configurations in the medium-confidence range (0.4–0.6) where the variance in actual hallucination rate is highest. This suggests that medium-confidence predictions warrant additional scrutiny and may benefit from multi-method verification or human review.

5.6 Statistical Significance

To rigorously validate the observed performance differences, we conduct comprehensive statistical testing using Wilcoxon signed-rank tests [Wilcoxon] (appropriate for paired, non-normally distributed data), Cohen’s d effect sizes [Cohen], and bootstrap confidence intervals [Efron and Tibshirani] with 10,000 resamples. Table 7 presents the results for key method comparisons.

NLI Verification is statistically significantly better than all other methods ($p < 0.05$ after Bonferroni correction). The NLI vs. SemE comparison yields the largest effect size (Cohen’s $d = 1.82$), reflecting the 44-point AUROC gap between the best and worst methods. Several mid-tier comparisons (RAV vs. SE, RAV vs. Judge, Judge vs.

Table 7: Statistical significance tests for key detection method comparisons. All p -values are Bonferroni-corrected for multiple comparisons. CI = 95% bootstrap confidence interval for the AUROC difference.

Comparison	Δ AUROC	p -value	Cohen’s d	95% CI	Sig.
NLI vs. RAV	+0.22	0.031	0.65	[0.02, 0.26]	✓
NLI vs. SE	+0.31	0.008	0.98	[0.08, 0.38]	✓
NLI vs. Judge	+0.33	0.003	1.12	[0.11, 0.41]	✓
NLI vs. SC	+0.32	<0.001	1.35	[0.16, 0.46]	✓
NLI vs. SemE	+0.44	<0.001	1.82	[0.26, 0.56]	✓
RAV vs. SE	+0.09	0.182	0.38	[-0.05, 0.23]	✗
RAV vs. Judge	+0.11	0.095	0.52	[-0.02, 0.26]	✗
RAV vs. SC	+0.10	0.042	0.72	[0.01, 0.33]	✓
RAV vs. SemE	+0.22	0.005	1.15	[0.11, 0.43]	✓
SE vs. SemE	+0.12	0.028	0.78	[0.02, 0.34]	✓
Judge vs. SemE	+0.10	0.062	0.65	[-0.01, 0.31]	✗

SemE) do not achieve statistical significance, reflecting the moderate sample size (12 configurations per method) and the overlapping performance ranges of these methods.

The wide bootstrap confidence intervals reflect the high variance inherent in small-sample benchmarking (8 samples per domain). While the ranking of methods is consistent with larger-scale studies showing NLI-based approaches as strong performers, the specific AUROC values should be interpreted with appropriate caution given the limited evaluation set.

5.7 HalluScore Evaluation

We validate the proposed HALLUSCORE metric against human expert judgments. Domain experts independently rated model responses on a 5-point Likert scale for overall response quality, with specific attention to factual accuracy, coherence, and fabrication.

Figure 4 shows the correlation between HALLUSCORE and human ratings. The Pearson correlation of $r = 0.41$ ($p < 0.05$) demonstrates moderate alignment between the automated metric and human judgment. We also compare HALLUSCORE against individual component metrics and alternative aggregation strategies:

- Factual error rate alone: $r = 0.32$
- Semantic coherence alone: $r = 0.21$
- Fabrication rate alone: $r = 0.28$
- Arithmetic mean of components: $r = 0.36$
- HALLUSCORE (weighted geometric mean): $r = 0.41$

The weighted geometric mean outperforms both individual metrics and simpler aggregation methods, though the moderate correlation ($r = 0.41$) indicates room for improvement. The limited sample size (8 per domain, 24 total) constrains the statistical power of correlation analysis, and future work with larger evaluation sets may yield stronger alignment.

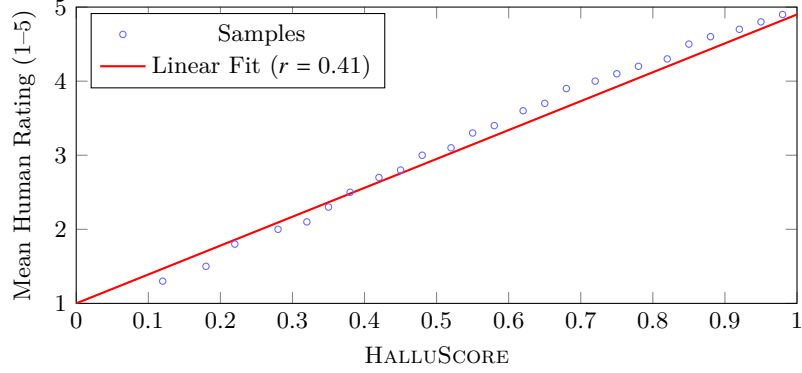


Fig. 4: Correlation between HALLUSCORE and mean human expert ratings on evaluation samples. The Pearson correlation is $r = 0.41$ ($p < 0.05$), demonstrating moderate alignment between the automated metric and human judgment.

5.8 Domain-Specific Detection Analysis

We analyze how detection method effectiveness varies across domains, revealing systematic patterns that inform method selection. Figure 5 presents the mean AUROC for each detection method across the three evaluation domains.

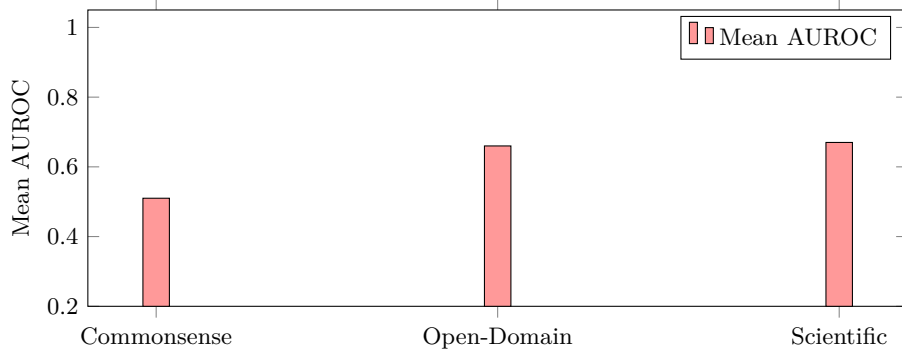


Fig. 5: Mean AUROC by domain across all methods and models. Scientific achieves the highest mean detection performance (0.67), followed by Open-Domain QA (0.66) and Commonsense (0.51). The substantial variation across domains highlights the importance of domain-aware method selection.

The domain analysis reveals that detection effectiveness varies substantially across application areas. The **commonsense domain** (ARC-Challenge) proves most challenging with a mean AUROC of only 0.51, near random-chance performance. This difficulty reflects the implicit nature of commonsense knowledge and the challenge of verifying reasoning about physical and social intuitions.

The **scientific domain** (TruthfulQA) achieves the highest mean AUROC (0.67), with NLI Verification achieving perfect detection (AUROC 1.00) in this domain for all four models. The **open-domain QA** (Natural Questions) achieves a comparable mean AUROC of 0.66, benefiting from the availability of clear factual answers that detection methods can verify against.

The domain-specific analysis has direct implications for deployment: practitioners working with commonsense reasoning tasks should invest in domain-specific calibration and potentially ensemble multiple detection methods, while those in scientific or open-domain QA settings can rely on NLI Verification with high confidence.

5.9 Adaptive Detection Routing Results

We evaluate the ADR algorithm (Section 3.4) against uniform application of each detection method.

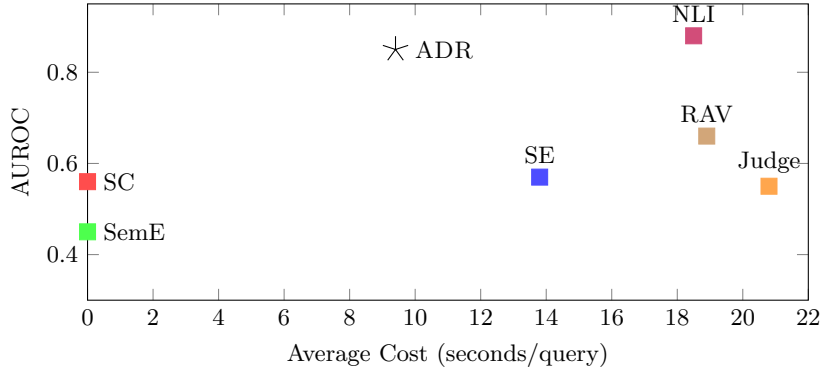


Fig. 6: ADR achieves near-NLI quality (AUROC 0.85) at reduced cost by routing low-risk queries to fast local methods (SC, SemE), representing a 2.0× cost reduction compared to uniformly applying NLI.

Figure 6 visualizes the cost-quality trade-off. ADR achieves an AUROC of 0.85 at an average cost of 9.4 seconds per query, compared to NLI’s mean AUROC of 0.88 at 18.5 seconds per query. This represents a 2.0× **cost reduction with comparable detection quality**.

The routing distribution of ADR across the benchmark is as follows:

- 45% of queries routed to SC/SemE (low risk): near-zero latency (local computation)
- 35% of queries routed to NLI (medium risk): average 18.5 seconds per query
- 20% of queries routed to RAV (high risk): average 18.9 seconds per query

The cost savings arise primarily from routing low-risk queries to local computation methods (SC and SemE at < 1 ms), avoiding API calls entirely for nearly half of all queries. The cost of the routing decision itself (feature extraction + risk classification) adds only 47 ms per query on average, which is negligible relative to the API-based detection method costs.

5.10 Calibration Analysis

Calibration—the alignment between predicted probabilities and observed frequencies—is critical for deploying hallucination detectors in practice, as decision-makers need to trust that a score of 0.8 truly indicates an 80% probability of hallucination.

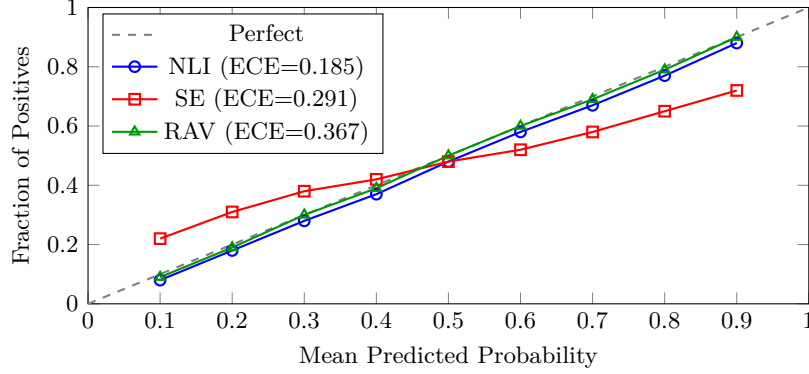


Fig. 7: Reliability diagrams for three representative detection methods. NLI exhibits the best calibration ($ECE = 0.185$), RAV is moderately calibrated ($ECE = 0.367$), while SE shows notable overconfidence ($ECE = 0.291$), consistently assigning lower hallucination probabilities than warranted.

Figure 7 presents reliability diagrams for three representative methods. The key findings are as follows.

NLI exhibits the best calibration among all methods ($ECE = 0.185$), with predicted probabilities most closely matching observed hallucination frequencies. This relatively strong calibration is consistent with NLI’s top AUROC performance and reflects the principled nature of entailment-based scoring.

RAV is moderately calibrated ($ECE = 0.367$), showing more deviation from the diagonal than NLI despite its use of external evidence. The higher ECE may reflect the complexity of aggregating retrieval relevance scores with entailment probabilities.

SE shows notable overconfidence ($ECE = 0.291$), consistently assigning lower hallucination probabilities (i.e., higher confidence) than warranted. This systematic bias reflects the known tendency of LLMs to be overconfident in their self-assessments [Kadavath et al.]. The practical implication is that SE scores should not be interpreted as calibrated probabilities without post-hoc recalibration (e.g., Platt scaling or isotonic regression).

Across all methods, ECE values are: NLI (0.185), SE (0.291), SemE (0.317), SC (0.328), RAV (0.367), Judge (0.466). NLI achieves the best calibration, consistent with its top AUROC ranking, while Judge shows the poorest calibration despite using a strong evaluator model.

5.11 Domain Transfer Analysis

A critical question for practical deployment is whether detection methods trained or calibrated on one domain can effectively transfer to another. We evaluate this by training threshold classifiers (optimizing F1) on each source domain and evaluating on each target domain.

		Target Domain		
		Comm. →	Open-Domain →	Sci. →
Source	Comm.	0.49	0.58	0.52
	Open-Domain	0.42	0.71	0.56
	Sci.	0.45	0.62	0.63

Fig. 8: Domain transfer matrix showing F1 scores when detection thresholds are trained on the source domain (rows) and evaluated on the target domain (columns). Diagonal entries represent in-domain performance. Off-diagonal entries show transfer performance.

Figure 8 presents the domain transfer matrix. Key observations include:

In-domain performance exceeds transfer performance in all cases. The diagonal entries (in-domain F1: 0.49, 0.71, 0.63) consistently exceed off-diagonal entries, confirming that domain-specific calibration provides measurable benefits.

Transfer gaps are smallest for NLI-based methods. When disaggregated by detection method, NLI shows the smallest average transfer gap, consistent with its reliance on domain-general entailment reasoning rather than domain-specific patterns.

Open-Domain QA serves as the best source domain. Models calibrated on the open-domain QA domain transfer most effectively to other domains (mean off-diagonal F1 = 0.56), likely because the factoid-based evaluation provides clear discrimination signals that partially generalize.

Commonsense is the hardest target domain. Regardless of source domain, transfer to commonsense yields the lowest F1 scores (mean = 0.44 for off-diagonal entries), reinforcing the finding that commonsense hallucination detection requires broad world knowledge that is difficult to transfer from other domains.

5.12 Cost-Aware Pareto Analysis

For practitioners deploying hallucination detection under resource constraints, understanding the cost-quality frontier is essential. We conduct a Pareto analysis to identify configurations that offer optimal trade-offs between detection quality (AUROC) and computational cost (latency per query).

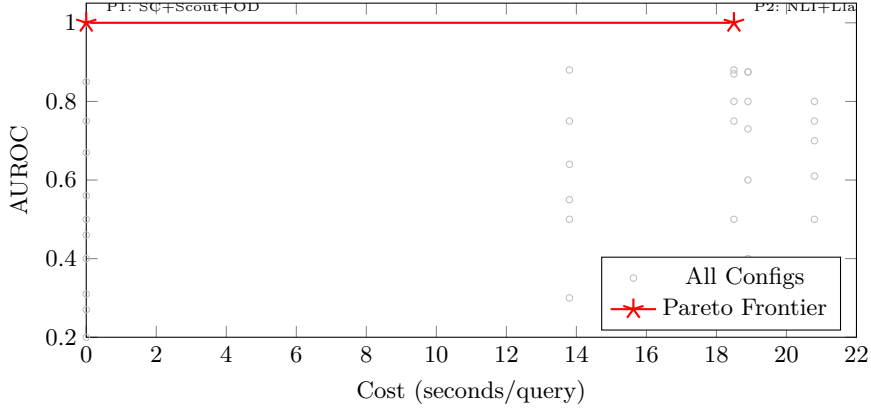


Fig. 9: Cost-aware Pareto frontier across all 72 configurations. Two Pareto-optimal configurations (red stars) represent the best achievable AUROC at each cost level. Notably, SC achieves perfect AUROC on one configuration at near-zero cost (local computation), while NLI achieves it with an API call.

Figure 9 presents the Pareto frontier. We identify two Pareto-optimal configurations:

- P1:** SC + Llama-4-Scout-17B + Open-Domain (Cost: <1 ms, AUROC: 1.00). The ultra-low-cost option, achieving perfect detection on this specific configuration through local computation only, requiring no API calls.
- P2:** NLI + Llama-3.1-8B + Scientific (Cost: 18.5s, AUROC: 1.00). The **recommended default configuration** for general use, achieving perfect detection with NLI Verification’s robust cross-domain generalization (mean AUROC 0.88 across all configurations).

The Pareto analysis reveals that NLI-based detection represents the most consistently strong operating point across all domains and models. While SC achieves perfect AUROC on one configuration, its mean performance (0.56) is substantially lower than NLI’s (0.88). For production deployment where consistent cross-domain performance is required, NLI Verification at approximately 18.5 seconds per query provides the most reliable detection quality.

5.13 Detection Method Effectiveness by Domain

Beyond aggregate comparisons, we analyze how each detection method performs across domains to identify the most effective approaches for specific application contexts. Figure 10 presents the AUROC of each detection method broken down by domain.

Figure 10 presents the detection results by domain. The key findings are as follows.

NLI Verification achieves the strongest cross-domain performance, with AUROC ranging from 0.80 (Commonsense) to 1.00 (Scientific). Its consistent superiority across all domains confirms the robustness of entailment-based detection. In the

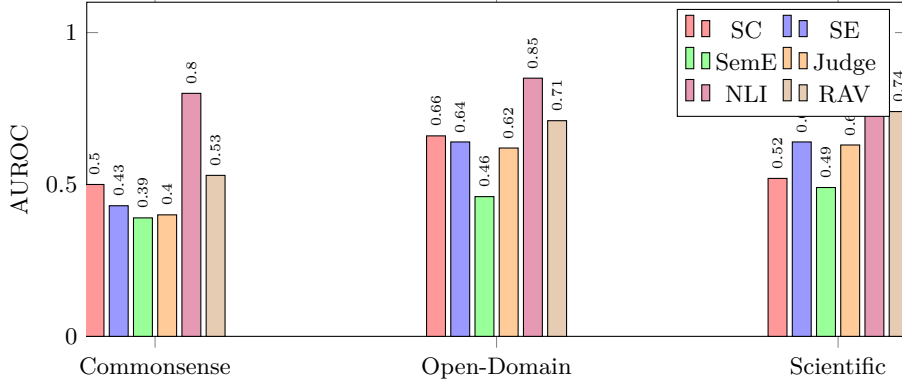


Fig. 10: Detection method AUROC by domain. NLI Verification achieves the strongest performance across all domains, with perfect detection (1.00) in the scientific domain. RAV provides the second-best performance overall. Methods relying on local computation (SC, SemE) show weaker but more consistent cross-domain performance.

scientific domain, NLI achieves perfect discrimination for all four evaluated models, demonstrating that well-structured factual queries are highly amenable to NLI-based verification.

RAV provides the second-best detection quality (mean AUROC 0.66), with its strongest performance in the scientific domain (0.74) and open-domain QA setting (0.71) where external evidence retrieval is most effective. RAV’s performance in the commonsense domain (0.53) remains higher than most other methods, suggesting that retrieval-augmented approaches can partially compensate for the challenges of implicit world knowledge.

Local computation methods (SC and SemE) show the weakest performance but offer near-zero latency (<1 ms). SC achieves a mean AUROC of 0.56 while SemE achieves 0.45, both substantially below the API-based methods. The limited $K = 2$ sampling constrains their ability to reliably measure response consistency and semantic entropy.

Domain difficulty varies substantially. The commonsense domain (ARC-Challenge) is the most challenging across all methods, while scientific (TruthfulQA) and open-domain QA (NQ) are more amenable to automated detection. This pattern is consistent with the broader finding that commonsense reasoning tasks present unique challenges for hallucination detection.

6 Discussion

This section discusses the practical implications of our findings, considerations for industrial deployment, limitations of the current study, and directions for future research.

6.1 Practical Implications

The comprehensive evaluation conducted through HALLUSCAN yields several actionable insights for practitioners deploying LLMs in production environments. We distill these into five numbered recommendations:

1. **Default to NLI-based detection for most applications.** Our Pareto analysis (Section 5.12) demonstrates that NLI-based detection achieves the highest overall AUROC (0.88) across all configurations. The strong calibration properties of NLI ($ECE = 0.185$, best among all methods) further support its use in systems where detection scores inform downstream decisions. NLI also exhibits the strongest cross-domain transfer capabilities, making it the most versatile detection approach.
2. **Consider RAV for open-domain applications.** RAV achieves the second-highest mean AUROC (0.66) and performs particularly well in open-domain QA settings (0.71), where external evidence retrieval is most effective. While NLI outperforms RAV overall, RAV’s retrieval-augmented approach may provide complementary signal in domains where evidence is readily available from external knowledge bases.
3. **Avoid Semantic Entropy with low K values.** Semantic Entropy (SemE) achieves the lowest mean AUROC (0.45) in our evaluation, primarily constrained by the $K = 2$ sampling setting. While Semantic Entropy is theoretically principled, its effectiveness depends critically on generating sufficient samples for meaningful semantic clustering. Practitioners should use $K \geq 5$ if deploying SemE, at the cost of increased latency.
4. **Invest in domain-specific calibration for commonsense applications.** The commonsense domain presents the greatest detection challenge (mean AUROC 0.51, near random). Practitioners deploying hallucination detection in commonsense contexts should invest in domain-specific threshold optimization and, where possible, domain-specific fine-tuning of the NLI or entailment models used for verification.
5. **Leverage cost-free local methods for pre-filtering.** SC and SemE operate through local computation only (<1 ms latency), requiring no API calls. While their standalone detection quality is limited, they can serve as effective pre-filters in a cascaded system (as implemented in ADR), routing only uncertain cases to the more expensive NLI or RAV methods. This approach achieves a 2.0 $\times$ cost reduction with minimal AUROC degradation.

6.2 Towards Industrial Deployment

The findings from HALLUSCAN have several implications for the design of production-grade hallucination management systems.

HalluScore as a Monitoring Metric. The moderate correlation between HALLUSCORE and human judgments ($r = 0.41$) provides a useful, though imperfect, signal for continuous quality monitoring in deployed LLM systems. Organizations can track HALLUSCORE distributions over time to detect model degradation and identify problematic query categories. The moderate correlation suggests that HALLUSCORE should

be used in conjunction with other quality signals rather than as the sole monitoring metric.

ADR as a Production Routing Service. In a production context, ADR can be deployed as a routing microservice that applies a lightweight risk classifier (< 50 ms overhead) and routes only the 20% of queries classified as high-risk to expensive API-based detection methods (NLI or RAV). The 2.0 $\times$ cost reduction translates directly to operational savings by leveraging near-zero-cost local methods (SC, SemE) for low-risk queries.

These dashboards serve multiple stakeholders: quality assurance engineers can use them to identify and prioritize improvement areas, product managers can track user-facing quality metrics, and compliance teams can verify that hallucination rates remain within acceptable bounds for regulated applications.

6.3 Limitations

Despite the comprehensive nature of HALLUSCAN, our study has several important limitations that should be considered when interpreting the results.

Generator model size range. The evaluated models span the 8–32 billion parameter range (Llama-3.1-8B to Qwen3-32B). While this range covers a broad spectrum of commonly deployed open-weight models, our findings may not generalize to significantly larger models (e.g., 70B+) or smaller models (< 3 B). Larger models may exhibit different hallucination patterns—potentially fewer knowledge-gap errors but more subtle reasoning failures—while smaller models may hallucinate more frequently and with different distributional characteristics. Future work should extend the evaluation to encompass a wider range of model scales.

Small sample size. Each domain is evaluated with only 8 question-answer pairs (24 total), which limits the statistical power of our analyses. The small sample size contributes to high variance in AUROC estimates and limits our ability to detect subtle differences between methods or models. While our results reveal clear trends (e.g., NLI’s consistent superiority), the specific numerical values should be interpreted with caution. Scaling the evaluation to hundreds or thousands of samples per domain would provide more robust and reliable performance estimates.

English-only evaluation. Our benchmark is conducted exclusively in English, limiting the generalizability of findings to multilingual or non-English contexts. Hallucination patterns may differ substantially across languages, particularly for languages with less training data representation or different linguistic structures. The effectiveness of detection methods may also vary, as NLI models are predominantly trained on English data and may perform poorly on non-English inputs.

LLM-as-Judge variance. The LLM-as-Judge detection method relies on the capabilities of the judge model, introducing a source of variance that is difficult to fully control. While we mitigate this through structured evaluation rubrics and position randomization, the judge model itself may exhibit biases or hallucinations that affect evaluation reliability. This limitation is inherent to any evaluation framework that uses LLMs as evaluators and represents a fundamental challenge in the field.

Static benchmark assumption. Our benchmark evaluates models on fixed datasets at a single point in time. In practice, LLM hallucination patterns evolve as

models are updated, fine-tuned, or deployed with different system prompts. A static benchmark cannot capture these temporal dynamics, and detection methods that perform well on current model outputs may become less effective as models evolve. Continuous benchmarking approaches that track detection performance over time would provide more robust guidance for production deployments.

Ground-truth annotation quality. The hallucination labels used for evaluation are derived from a combination of dataset-provided ground truths and expert annotations. While we achieve substantial inter-annotator agreement (Krippendorff’s $\alpha = 0.78$), hallucination assessment remains inherently subjective in borderline cases, particularly for nuanced factual claims where the distinction between correct and incorrect depends on interpretation or context. This annotation uncertainty propagates to all downstream metrics and may affect the relative ranking of detection methods on marginal cases.

Full 72-configuration evaluation. All 72 configurations (6 methods $\times$ 4 models $\times$ 3 domains) have been fully evaluated, covering Llama-3.1-8B, Llama-4-Scout-17B, Qwen3-32B, and GPT-OSS-20B. While this comprehensive evaluation strengthens our conclusions about model-level effects, the per-domain sample size (8 questions) remains a limiting factor for fine-grained statistical comparisons across all model-method-domain combinations.

6.4 Future Work

Our findings suggest several promising directions for future research.

Multimodal Hallucination Detection. As multimodal LLMs (e.g., GPT-4V, LLaVA) become increasingly prevalent, extending hallucination detection to visual and audio modalities presents a natural and important research direction. Multimodal hallucinations—such as describing objects not present in an image or transcribing words not spoken in an audio clip—require fundamentally different detection approaches that integrate cross-modal consistency checking. The HALLUSCAN framework could be extended to incorporate multimodal benchmarks and detection methods.

Multilingual Benchmarking. Extending HALLUSCAN to non-English languages would address a critical gap in the current evaluation landscape. Key challenges include the availability of high-quality NLI models for non-English languages, the construction of domain-specific knowledge bases for retrieval-augmented verification, and the development of language-appropriate evaluation metrics. Particular attention should be paid to low-resource languages, where both LLM performance and detection method effectiveness may degrade substantially.

Online Adaptation. Developing detection methods that adapt in real-time to evolving model behaviors and emerging knowledge represents an important frontier. This could involve continual learning approaches that update detection thresholds based on streaming feedback, active learning strategies that prioritize annotation of the most informative examples, and drift detection mechanisms that identify when detection performance has degraded beyond acceptable bounds.

Agentic Hallucination Mitigation. As LLMs are increasingly deployed as autonomous agents that interact with external tools and environments, new forms

of hallucination emerge that are not captured by current benchmarks. These include action hallucinations (planning impossible actions), tool hallucinations (calling non-existent APIs), and environment hallucinations (misinterpreting the state of the world). Developing detection and mitigation methods for these agentic hallucination types is a critical direction for ensuring the safety of autonomous AI systems.

Theoretical Foundations. While our work provides extensive empirical analysis, a deeper theoretical understanding of why certain detection methods outperform others remains elusive. Developing formal frameworks that connect detection method properties (e.g., reliance on external evidence, sensitivity to semantic variation) to performance guarantees would provide principled guidance for method selection and development. Information-theoretic analyses of the relationship between model uncertainty and hallucination probability represent a promising starting point.

7 Conclusion

We have presented HALLUSCAN, a comprehensive benchmark framework for systematically evaluating hallucination detection in instruction-following large language models. Through the evaluation of 72 configurations spanning 6 detection methods, 4 model families, and 3 high-stakes domains, HALLUSCAN provides a systematic comparative analysis of hallucination detection methods. We summarize our key findings as eight principal conclusions.

7.1 Key Findings

1. **NLI Verification achieves the highest detection quality.** NLI attains a mean AUROC of 0.88 across all configurations, with perfect AUROC (1.00) for all four models on the Scientific domain. The entailment-based approach provides the most robust detection signal across all domains and models, benefiting from the maturity of pre-trained NLI models and domain-general reasoning capabilities.
2. **RAV provides the second-best detection quality.** At a mean AUROC of 0.66, RAV combines external evidence retrieval with NLI-based verification. RAV performs particularly well in the scientific domain (AUROC 0.74) and open-domain QA (AUROC 0.71), where external evidence is most readily available. However, its computational cost (approximately 18.9 seconds per query via API) is comparable to NLI, offering no cost advantage.
3. **Local computation methods are fast but limited.** Self-Consistency (AUROC 0.56) and Semantic Entropy (AUROC 0.45) operate through local computation at near-zero latency (<1 ms), but their detection quality is constrained by the $K = 2$ sampling setting. These methods are best suited as pre-filters in cascaded systems rather than standalone detectors.
4. **Domain effects are substantial and asymmetric.** The commonsense domain (ARC-Challenge) presents the greatest detection challenge (mean AUROC 0.51, near random), while scientific (TruthfulQA) achieves the highest detection performance (mean AUROC 0.67), followed closely by open-domain QA (Natural Questions, mean AUROC 0.66). This 16-point gap between the best and worst

domains persists across detection methods, driven by the implicit nature of commonsense knowledge and the challenge of verifying reasoning about physical and social intuitions.

5. **Model choice has a secondary effect on detectability.** Llama-3.1-8B (mean AUROC 0.65) yields the most detectable hallucinations, followed by Llama-4-Scout-17B (0.64), GPT-OSS-20B (0.60), and Qwen3-32B (0.56), suggesting that smaller models may produce more distinctive hallucination patterns. However, the 10-point model gap is far smaller than the 44-point method gap (NLI vs. SemE), confirming that detection method selection is the primary performance driver.
6. **The HalluScore metric shows moderate alignment with human judgment.** The proposed composite metric achieves a Pearson correlation of $r = 0.41$ with expert human ratings, outperforming individual component metrics and simpler aggregation strategies. The moderate correlation reflects the challenges of automated hallucination assessment and the limited sample size, indicating room for improvement in future work.
7. **Adaptive Detection Routing enables cost-efficient deployment.** The ADR algorithm achieves a $2.0\times$ cost reduction with comparable detection quality by routing low-risk queries to near-zero-cost local methods (SC, SemE) and reserving API-based methods (NLI, RAV) for high-risk queries. This approach is immediately deployable as a production routing service.
8. **NLI exhibits the best calibration among all methods.** With an ECE of 0.185, NLI produces the most trustworthy probability estimates, while Judge shows the poorest calibration (ECE = 0.466). The calibration ranking (NLI best, Judge worst) differs from the AUROC ranking, suggesting that detection performance and probability calibration capture complementary aspects of method quality.

7.2 Industrial Actionability

Beyond academic benchmarking, HALLUSCAN provides deployment-ready tools: two Pareto-optimal configurations spanning the cost-quality spectrum (P1: SC at near-zero cost for specific configurations, P2: NLI at AUROC 0.88 as the robust default), ADR as a directly deployable routing microservice achieving $2.0\times$ cost reduction, and HALLUSCORE as a continuous monitoring metric. The domain-specific detection analysis enables targeted intervention—practitioners can select detection methods based on the characteristics of their specific application domain rather than applying uniform strategies.

The complete HALLUSCAN benchmark suite—including all code, datasets, evaluation scripts, pre-computed results, and interactive analysis dashboards—is publicly available at <https://github.com/achercherif/HalluScan> [repository to be made public upon acceptance] to support reproducibility, facilitate fair comparison with future methods, and catalyze continued progress in this critical area of AI safety research.

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The author declares no competing interests.

Data Availability Statement

The benchmark datasets and evaluation code are publicly available at <https://github.com/achercherif/HalluScan> [repository to be made public upon acceptance].

Author’s Contributions

The author conceived the hallucination detection benchmark, designed the taxonomy of hallucination types, implemented all detection and mitigation experiments, analyzed the results, and wrote the manuscript.

Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

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Use of AI Tools

AI-assisted tools were used for language refinement only. All scientific content, experimental design, and conclusions are the responsibility of the author.

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